

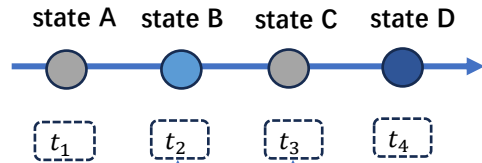
TSSMRBench Overview

Benchmarking whether memory systems can retrieve the correct temporal semantic state from evolving memories

1. Problem

Memories evolve over time.
Queries may target a non-latest state.

Evolution of a semantic entity



Which state
was valid at the
target stage?

! Latest is not always correct.

Real-world applications



Legal revision lookup

Retrieve the law version valid
at the time of a past case.



Plan-version comparison

Compare a previously proposed
plan with the current plan.



Financial state ordering

Order historical market states to
support investment reasoning

2. Benchmark Design

Three task types with different
supporting state sizes.

Single-State Lookup

support size $|G(q)| = 1$



Cross-Version Comparison

support size $|G(q)| = 2$



Temporal Ordering

support size $|G(q)| = 3-5$



Design Principles



state-
specific



multi-
version



evidence-
grounded

3. Dataset Construction

Transform raw evolution records into
structured temporal semantic states
and diverse QA



Step A Source materials

Collect official release notes as
evolution memories



Step B 30-state evolution window

Select the latest 30 release notes
as semantic state versions.



Step C LLM rewriting

Use deepseek-v4-pro rewrite raw
records into clean, noise-removed,
natural-language memory units.



Step D QA generation.

Generate 3 multiple-choice questions
per repository across the three task
types.



300
items



Memory nodes
(30 states×300)

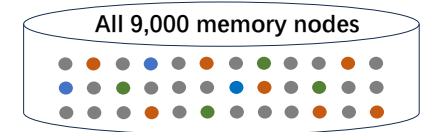


900
questions

4. Evaluation Protocol

All memories are mixed into one pool.
Systems answer the same questions
under unified settings.

Shared Mixed Memory Pool



Memory systems



Oracle
Gold
Context



BM25



FAISS



Mem0



Graphiti

Retrieve Top-10 (by each system)



Answer Model (deepseek-v4-pro)
Reads retrieved memories (top-k)
and produces final answers

answers

Evaluation Metrics



ACC
Accuracy



Cov
Coverage



CSR
Correct-
Support Rate



Latency
Retrieval
Latency